Student ID: 24IT087

Chandubhai S Patel Institute of Technology

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
B. TECH (IT/CE/CSE/AIML)

DEPSTAR

Subject: Advanced Web Development Frameworks (ITUE301) | Semester: 5th

Practical Statement:

- Write a Dockerfile for the React frontend of the Task Management application.
- Write a Dockerfile for the Node/Express backend.
- Create a docker-compose.yml file that orchestrates all 3 services: frontend, backend, and MongoDB.
- Configure correct port mappings and an internal Docker network so the services can communicate.
- The entire application must start with a single docker-compose up command, with no manual steps in between.

Screenshots:

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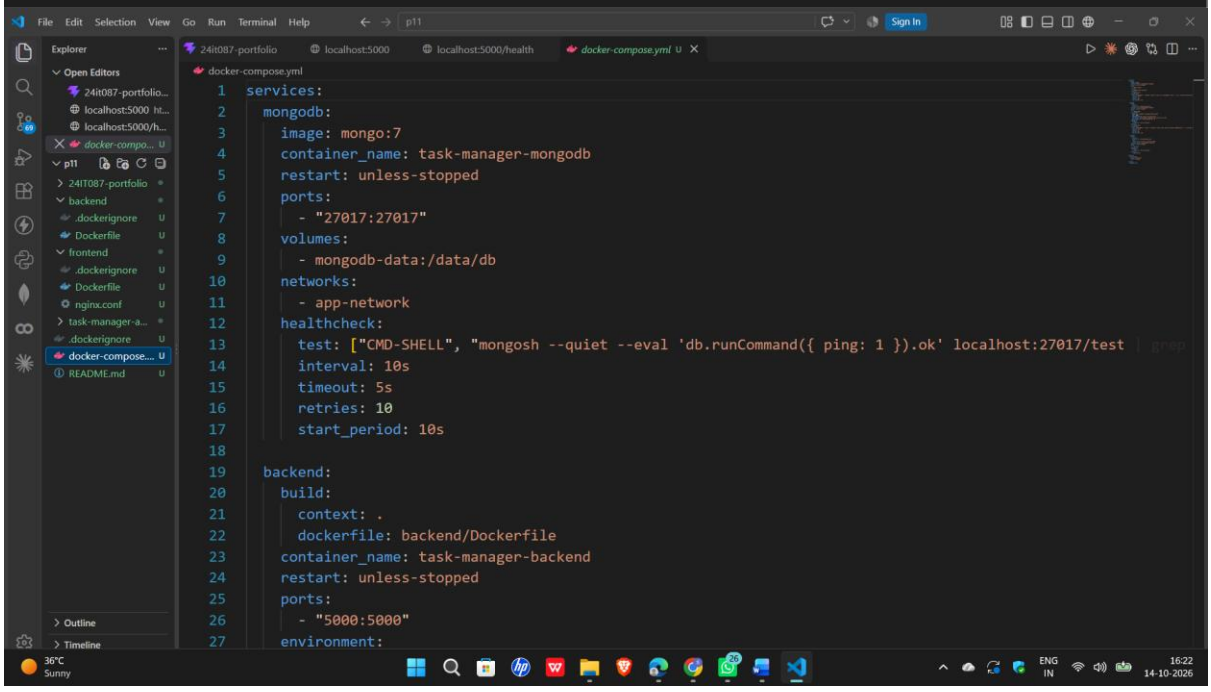
```
PS D:\sem 5\ITUE301_AWDF\p11> docker --version
Docker version 29.7.2, build a7dcaa6
PS D:\sem 5\ITUE301_AWDF\p11> docker compose version
Docker Compose version v5.3.1
PS D:\sem 5\ITUE301_AWDF\p11> docker info
Client:
 Version:    29.7.2
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.119.0
    Path:     C:\Users\Digisha\.docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.30.0
    Path:     C:\Users\Digisha\.docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.36.0-desktop.1
    Path:     C:\Users\Digisha\.docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.3.1
    Path:     C:\Users\Digisha\.docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     C:\Users\Digisha\.docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.4.3
    Path:     C:\Users\Digisha\.docker\cli-plugins\docker-desktop.exe
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.7
    Path:     C:\Users\Digisha\.docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     C:\Users\Digisha\.docker\cli-plugins\docker-extension.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
```

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```
PS D:\sem 5\ITUE301_AWDF\p11> docker info
PS D:\sem 5\ITUE301_AWDF\p11> docker compose config
name: p11
services:
  backend:
    build:
      context: D:\sem 5\ITUE301_AWDF\p11
      dockerfile: backend/Dockerfile
    container_name: task-manager-backend
    depends_on:
      mongodb:
        condition: service_healthy
        required: true
    environment:
      FRONTEND_ORIGIN: http://localhost:5173
      JWT_EXPIRES_IN: 1h
      JWT_SECRET: change-this-container-secret
      MONGO_URI: mongodb://mongodb:27017/taskdb
      PORT: "5000"
    healthcheck:
      test:
        - CMD-SHELL
        - 'node -e "require(''http'').get(''http://localhost:5000/health'', r => process.exit(r.statusCode
          === 200 ? 0 : 1)).on(''error'', () => process.exit(1))"'
      timeout: 5s
      interval: 10s
      retries: 10
      start_period: 10s
    networks:
      app-network: null
    ports:
      - mode: ingress
        target: 5000
        published: "5000"
        protocol: tcp
    restart: unless-stopped
  frontend:
```

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```
  frontend:
    build:
      context: D:\sem 5\ITUE301_AWDF\p11
      dockerfile: frontend/Dockerfile
      args:
        VITE_API_URL: http://localhost:5000
    container_name: task-manager-frontend
    depends_on:
      backend:
        condition: service_healthy
        required: true
    networks:
      app-network: null
    ports:
      - mode: ingress
        target: 80
        published: "5173"
        protocol: tcp
    restart: unless-stopped
  mongodb:
    container_name: task-manager-mongodb
    healthcheck:
      test:
        - CMD-SHELL
        - 'mongosh --quiet --eval ''db.runCommand({ ping: 1 }).ok'' localhost:27017/test
          | grep 1'
      timeout: 5s
      interval: 10s
      retries: 10
      start_period: 10s
    image: mongo:7
    networks:
      app-network: null
    ports:
      - mode: ingress
        target: 27017
```



Github Repo Link:
https://github.com/digishasavaliya/ITUE301_AWDF.git